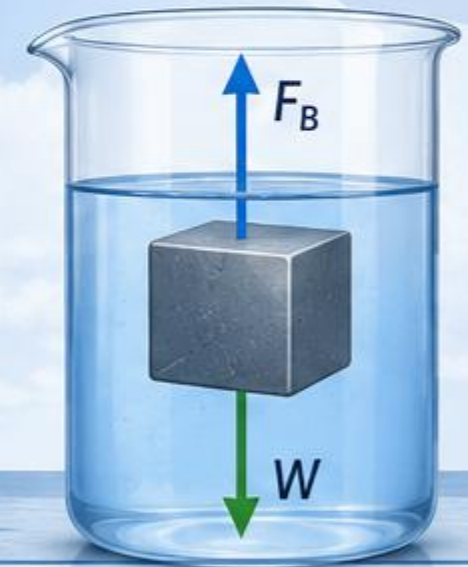
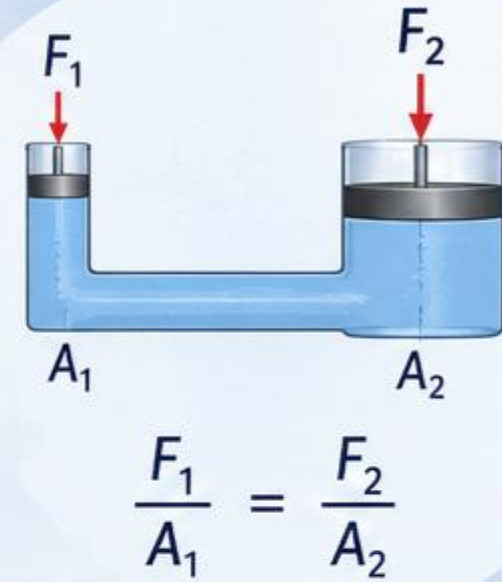


Applications of Pascal's Principle — and — Archimedes' Principle

General Science 11



$$F_B = \rho_{\text{fluid}} g V_{\text{displaced}}$$

F_B = buoyant force

ρ_{fluid} = density of the fluid

$V_{\text{displaced}}$ = volume of fluid displaced by the object

Presented by:



Learning Objectives

At the end of this lesson, you should be able to:



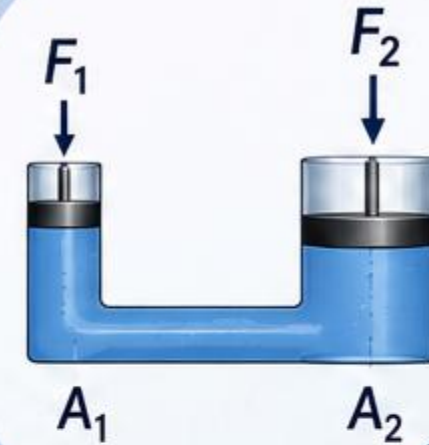
- **Explain** Archimedes' Principle and Pascal's Principle.



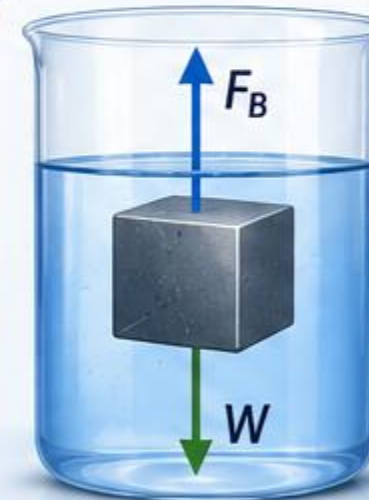
- **Identify** real-life applications of both principles in homes, communities, businesses, and transportation.



- **Recognize** the importance of fluid principles in solving everyday problems.



Pascal's Principle



Archimedes' Principle





Fluid mechanics

is the study of how **liquids** and **gases** behave and respond to forces.

Its principles are applied in:



Engineering



Transportation



Medicine



Construction



Manufacturing



Everyday technologies



Key Question

How do hydraulic machines lift heavy objects?



Why do ships float while stones sink?



Two Important Fluid Principles

Fluid Principles

Two important principles explain the behavior of fluids:



Pascal's Principle

Explains how pressure is transmitted through confined fluids.

Example: Hydraulic Jack



$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

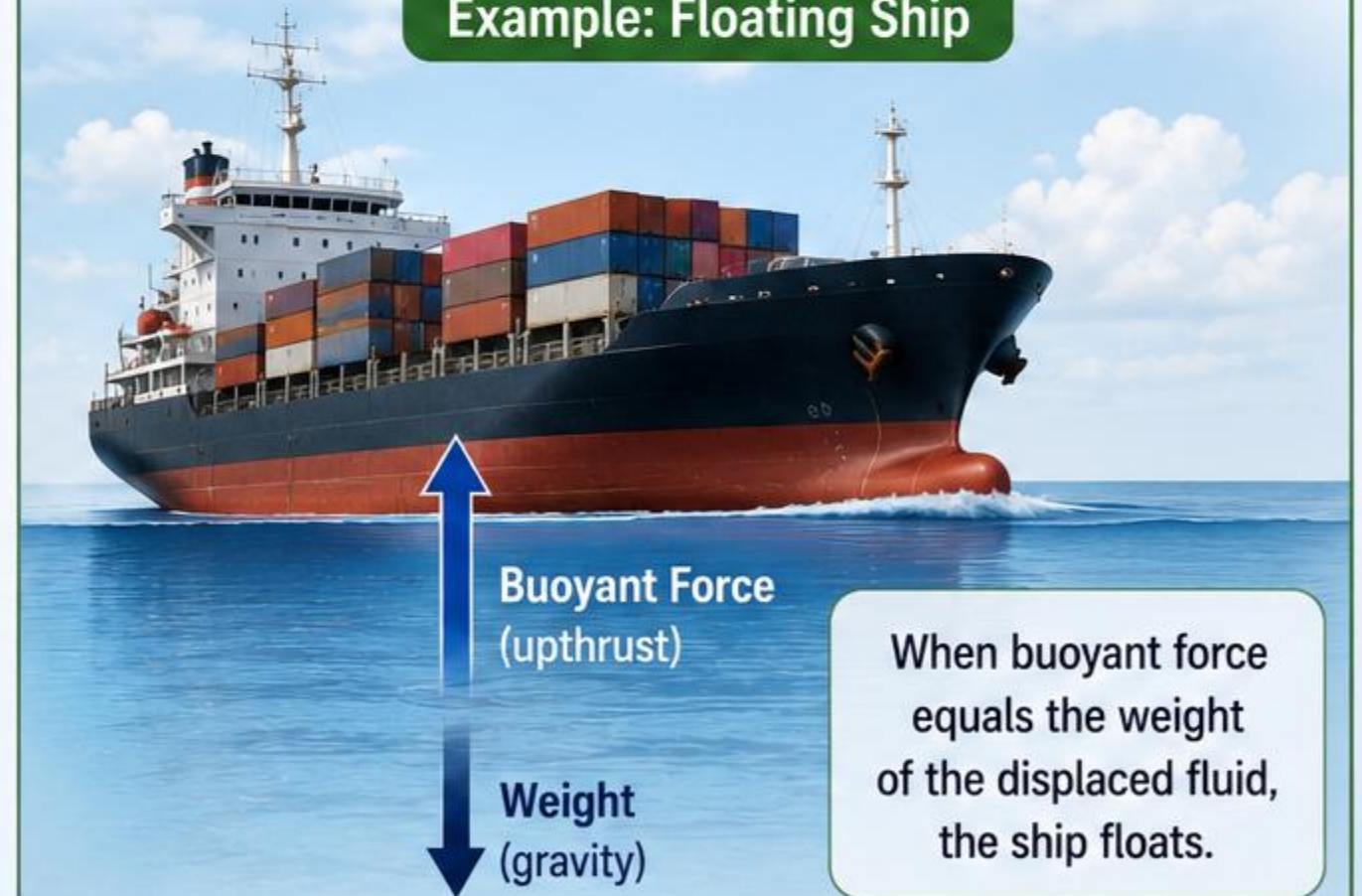
Pressure is transmitted equally throughout the fluid.



Archimedes' Principle

Explains why objects float or sink in fluids.

Example: Floating Ship





Pascal's Principle

Pascal's Principle

French mathematician
Blaise Pascal discovered that:

“ Pressure applied to a confined fluid is transmitted **equally** and **undiminished** throughout the entire fluid. ”

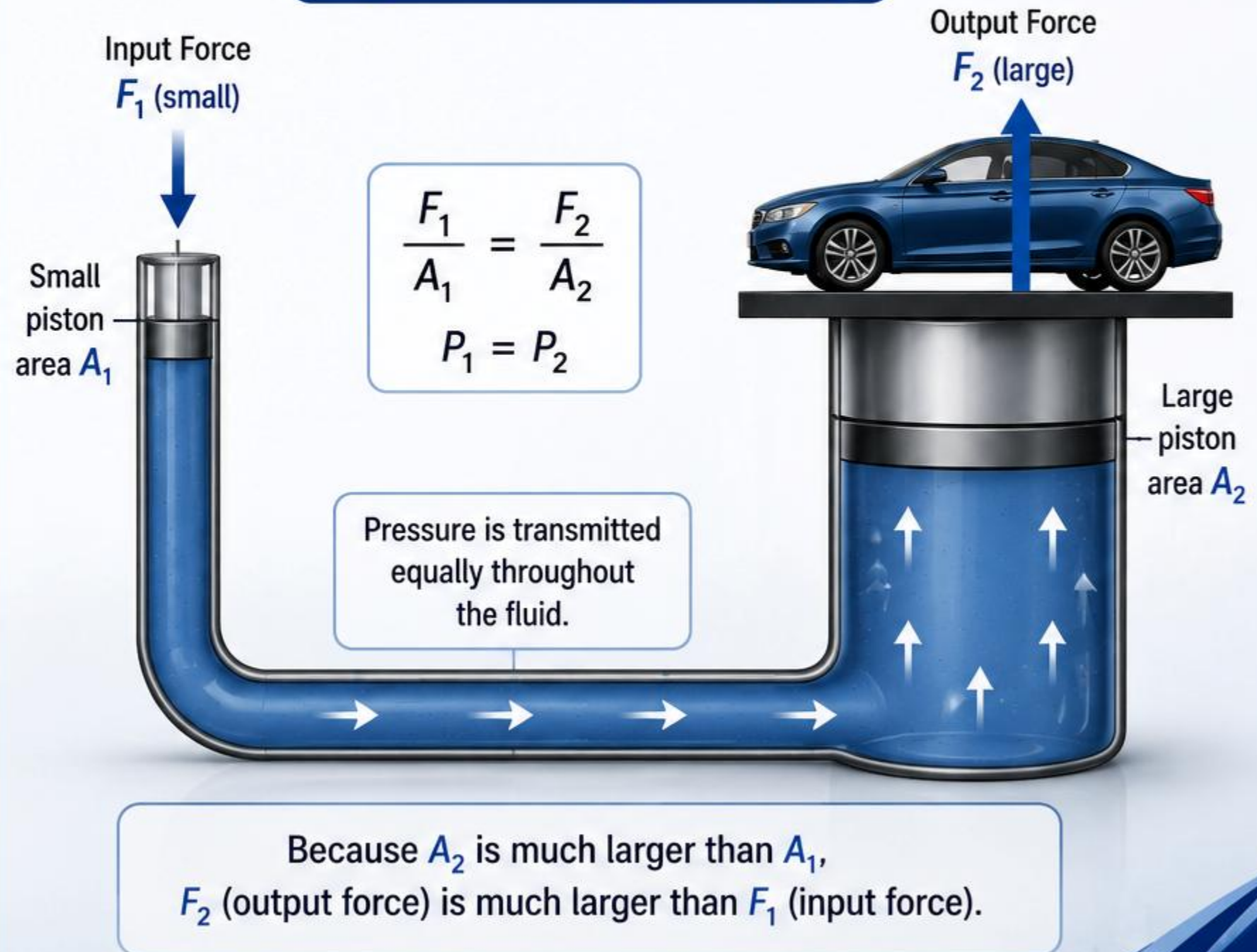


Key Idea



A **small force** can produce a **much larger force** using hydraulic systems.

Hydraulic Lift (Example)



Pascal's Principle Formula

Mathematical Relationship

Pressure

$$P = \frac{F}{A}$$

Since pressure is equal throughout the hydraulic system,

$$P_1 = P_2$$

Therefore,

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

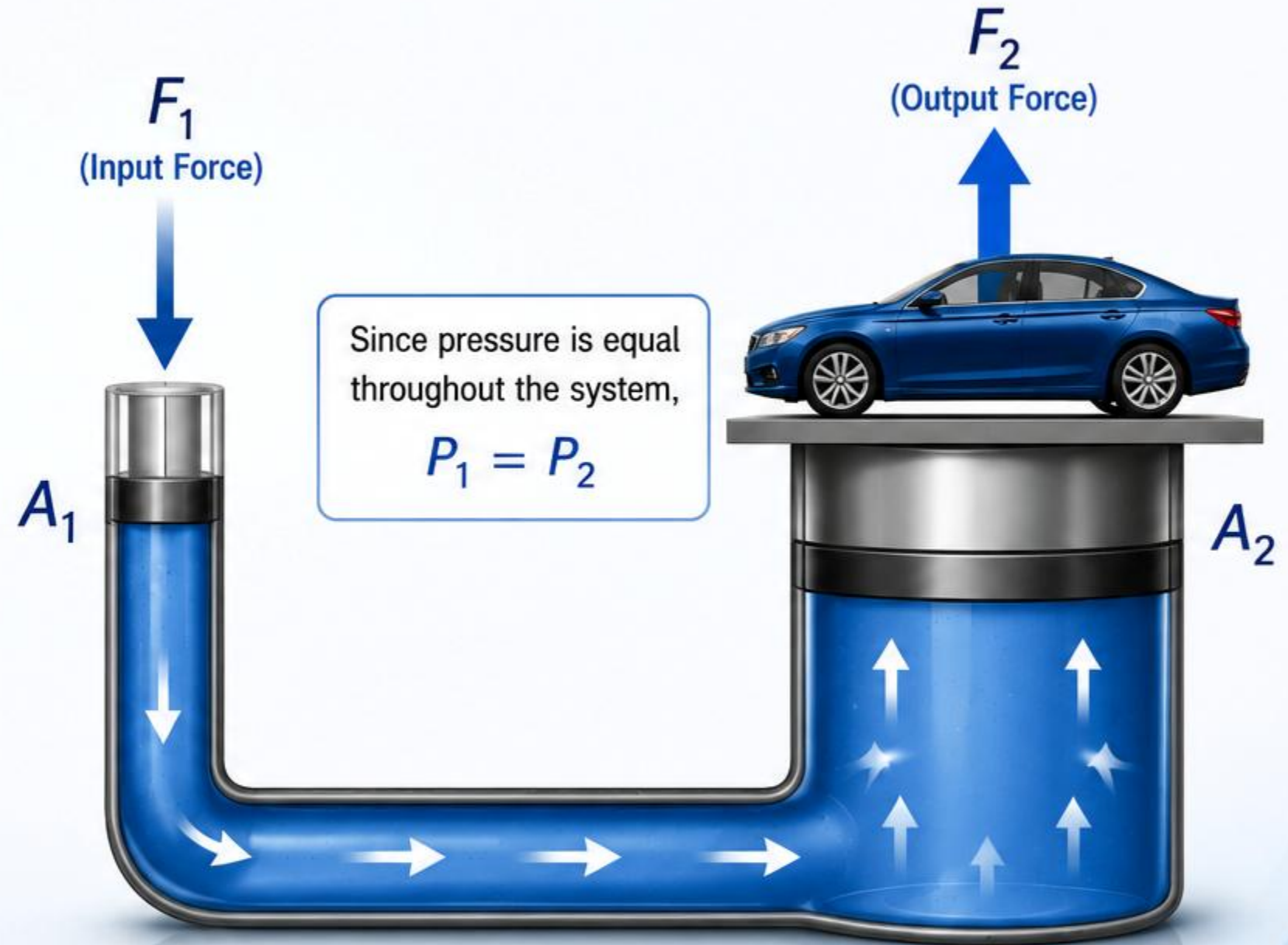
Where:

- F_1 = Input Force
- A_1 = Small piston area
- F_2 = Output Force
- A_2 = Large piston area



Key Idea

A **small** input force can generate a **large** output force.



Because A_2 is much larger than A_1 , F_2 (output force) is much larger than F_1 (input force).





Archimedes' Principle

Archimedes' Principle

Greek mathematician **Archimedes** stated:

“An object partially or completely immersed in a fluid experiences an upward **buoyant force** equal to the **weight of the fluid it displaces**.”



Key Idea



Objects float because of **buoyant force**.



Buoyant Force (F_b)

Upward force exerted by the fluid

Weight (W)

Downward force due to gravity

The boat displaces water whose weight equals the buoyant force.

Archimedes' Principle

$$F_b = W_{\text{fluid displaced}}$$

$$\text{Buoyant force} = \text{Weight of the fluid displaced}$$

Buoyant Force

What is Buoyant Force?



Buoyant force is the **upward force** exerted by a fluid that **opposes the weight** of an immersed object.

Whether an object floats or sinks depends on the relationship between:

Weight

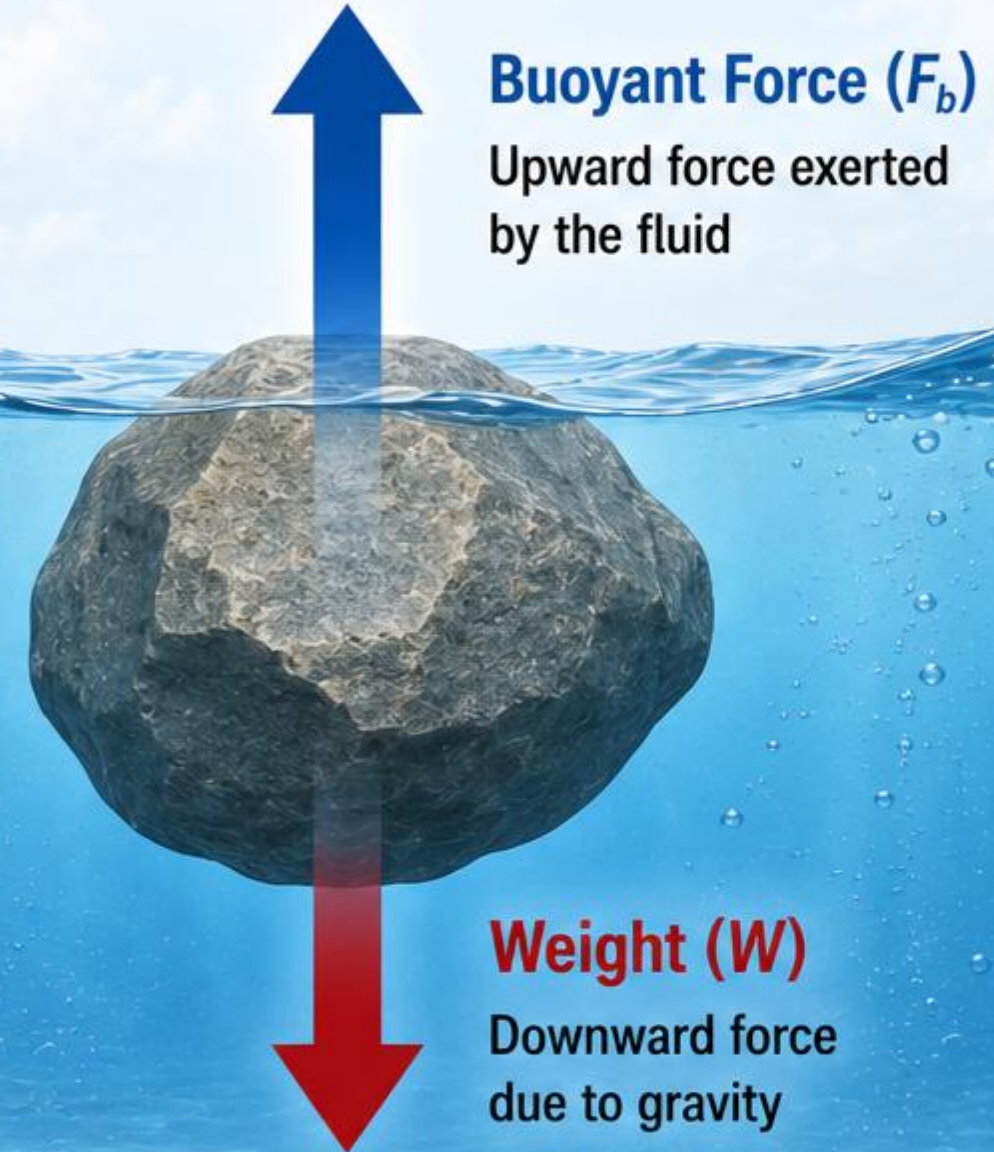


&

Buoyant Force



Condition		Relationship	Result
Floats		Buoyant Force > Weight	Object floats
Neutral Float		Buoyant Force = Weight	Object remains suspended
Sinks		Buoyant Force < Weight	Object sinks



The greater the **buoyant force**, the more likely the object will **float**.

Floating, Sinking, and Neutral Buoyancy

FLOATING

Buoyant Force $\geq$ Weight



Examples:

- Boats
- Cargo ships
- Life jackets



NEUTRAL BUOYANCY

Buoyant Force = Weight



Examples:

- Scuba divers (hovering)
- Submarines (neutrally balanced)
- Underwater robots



SINKING

Buoyant Force $<$ Weight



Examples:

- Stones
- Metal objects
- Most dense materials



The behavior of an object in a fluid depends on the relationship between the **buoyant force** and its **weight**.

Factors Affecting Buoyancy

What Determines if an Object Floats?

1 Shape

A wider or hollow shape displaces more fluid.



Key Idea

More displaced fluid means greater **buoyant force**.

Example:

- Clay ball → **sinks**
- Clay boat → **floats**

Clay ball



Displaces less fluid.
 $\text{Buoyant force} < \text{Weight}$

SINKS

Clay boat



Displaces more fluid.
 $\text{Buoyant force} > \text{Weight}$

FLOATS

Why Shape Matters



Objects with shapes that push away (**displace**) more fluid experience greater **buoyant force**, making them more likely to float.

Factors Affecting Buoyancy

2

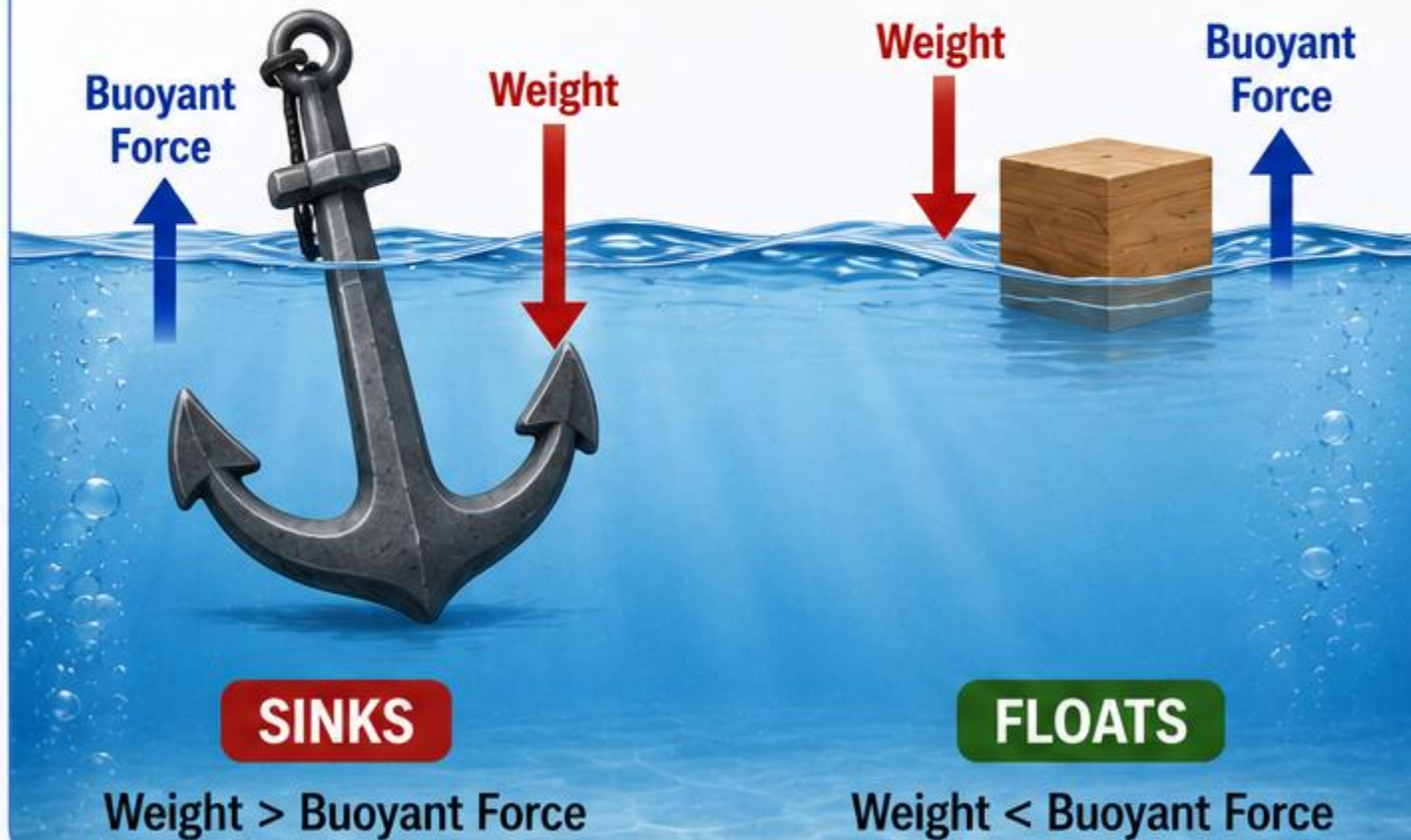
Mass



Greater mass generally increases weight, making **sinking** more likely if buoyant force is **insufficient**.

Greater Mass

Lesser Mass



3

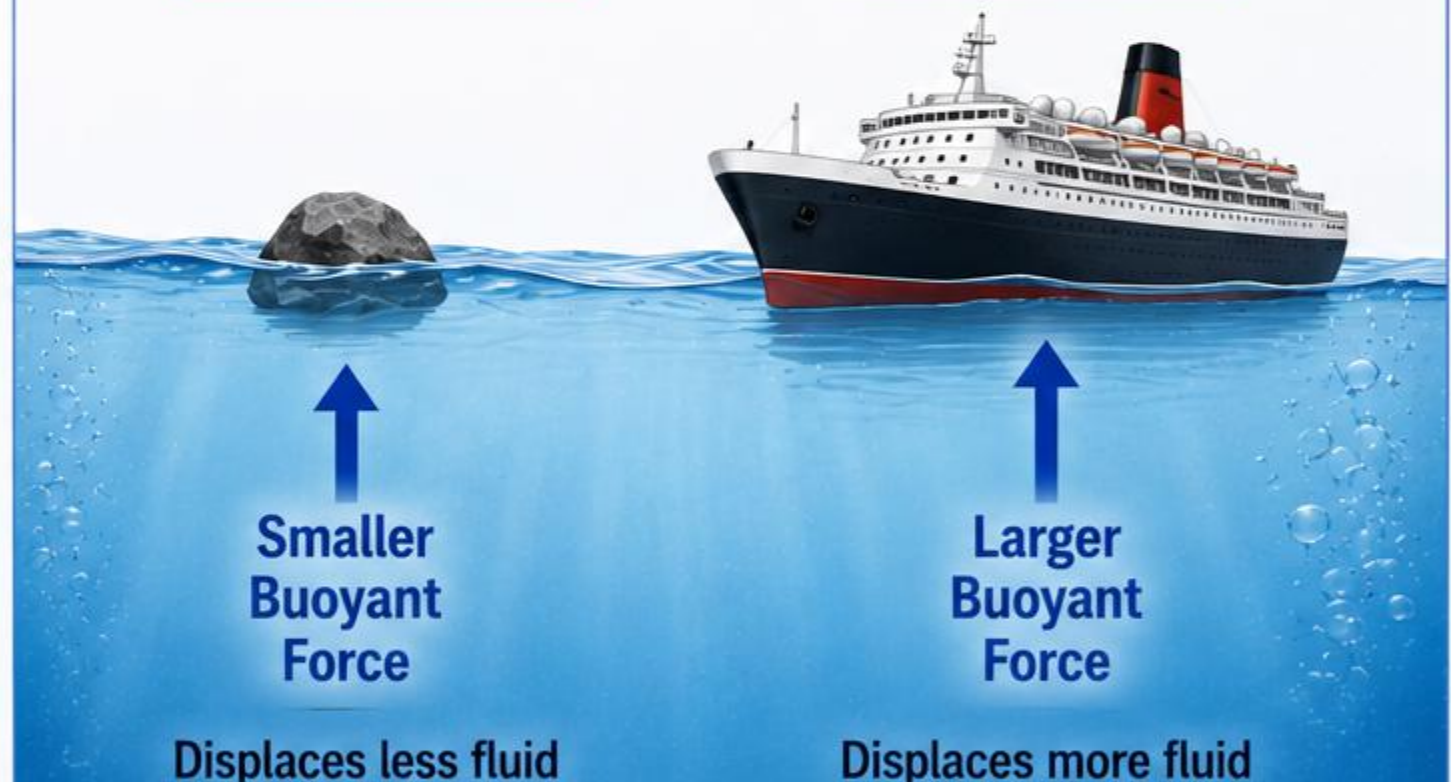
Volume



Larger volume displaces more fluid, producing **greater buoyant force**.

Smaller Volume

Larger Volume



Key Takeaway

More **mass** increases weight.

More **volume** increases displaced fluid.

Greater displaced fluid
means greater **buoyant force**.

EASY WAY TO REMEMBER



MASS =

How much **stuff** is in an object.



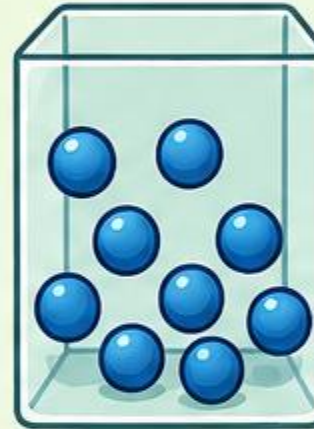
WEIGHT =

How hard **gravity** pulls on that object.

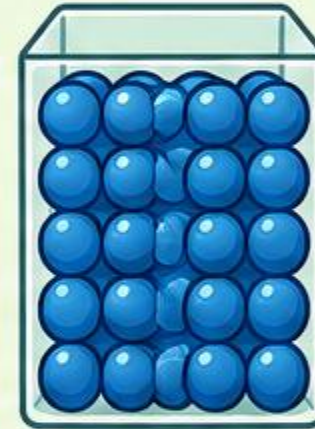


DENSITY =

How **tightly** the stuff is packed together.



Less dense
(loosely packed)

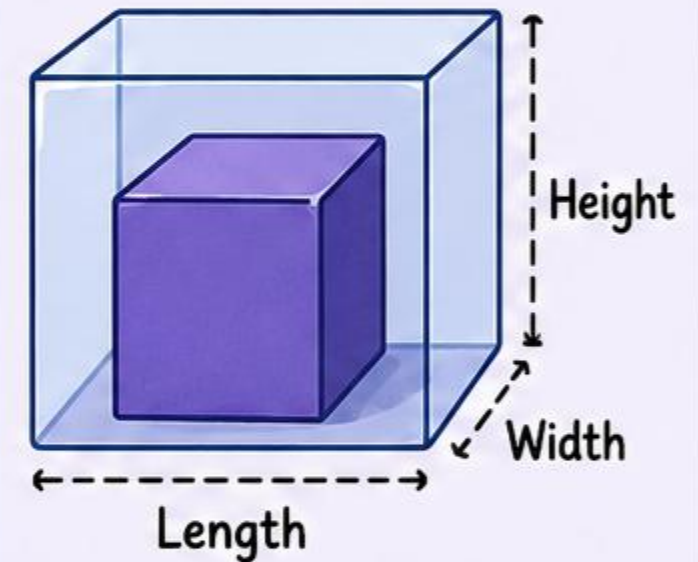


More dense
(tightly packed)



VOLUME =

tells us how much **space** an object takes up.



- **Mass** is always the **same**.
- **Weight** can change.
- **Density** depends on how it's packed.
- **Volume** is the space it takes.



Remember:

Mass is what's in.
Weight is gravity's pull.
Density is how it's packed.
Volume is space it fills!



Easy Way to Remember



Mass

= How much stuff is in an object.

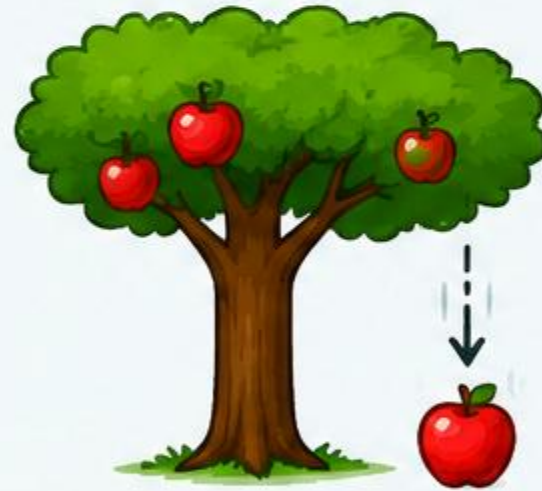


More marbles means more **mass**.



Weight

= How hard gravity pulls on that object.

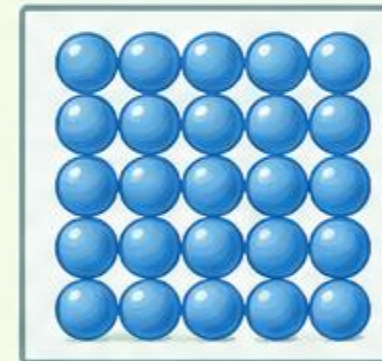


Gravity pulls the apple down. That pull is **weight**.

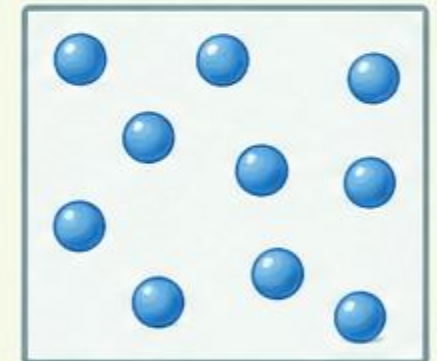


Density

= How tightly the stuff is packed together.



High density



Low density

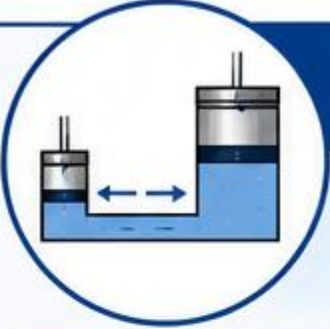
More packed = **higher density**.
Less packed = **lower density**.

★ In short: **Mass** is about **amount**. | **Weight** is about **gravity**. | **Density** is about **packing**.



Real-Life Applications

Applications in Homes and Communities



PASCAL'S PRINCIPLE

Pressure applied to a confined fluid is transmitted equally and undiminished throughout the entire fluid.



Hydraulic office chair



Hydraulic bottle jack



Garbage truck



Fire truck ladder



ARCHIMEDES' PRINCIPLE

An object partially or completely immersed in a fluid experiences an upward buoyant force.



Life jacket



Rescue boat



Floating dock



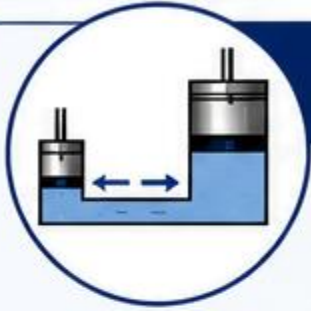
Inflatable swimming tube



These everyday applications show how fluid mechanics principles make our lives **safer**, **easier**, and **more efficient**.

Applications in Transportation

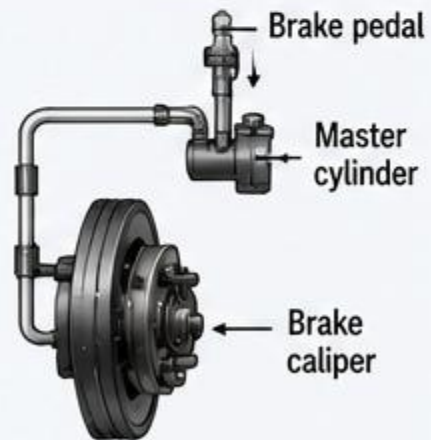
TRANSPORTATION



PASCAL'S PRINCIPLE

Pressure applied to a confined fluid is transmitted equally and undiminished throughout the fluid.

Hydraulic brake systems



Hydraulic steering



Aircraft landing gear



ARCHIMEDES' PRINCIPLE

An object partially or completely immersed in a fluid experiences an upward buoyant force.

Ships



Boats



Submarines



Floating bridges



Hot-air balloons



WHY ARE THESE IMPORTANT?

These technologies improve transportation **efficiency**, **safety**, and **reliability**.

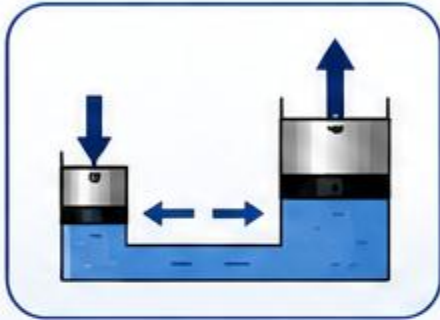
COMPARISON

Pascal's Principle vs. Archimedes' Principle

PASCAL'S PRINCIPLE

VS.

ARCHIMEDES' PRINCIPLE



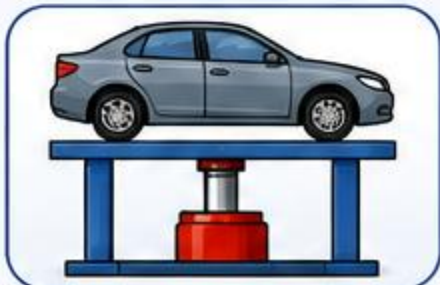
Pressure transmission



Confined fluids



Hydraulic systems



Multiplies force



Buoyancy



Immersed objects



Floating and sinking



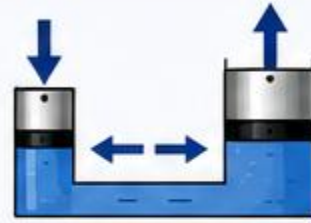
Explains buoyant force

KEY TAKEAWAYS

Remember!



Pascal's Principle explains pressure transmission in confined fluids.



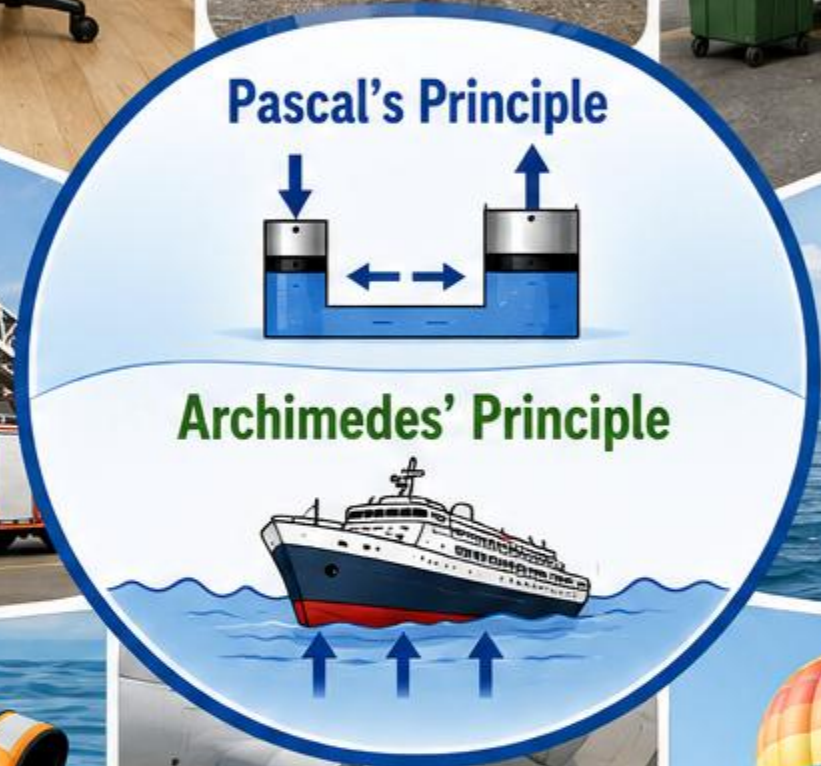
Archimedes' Principle explains buoyant force and why objects float or sink.



Shape, mass, and volume affect buoyancy.



These principles are applied in **homes**, **communities**, **transportation**, **engineering**, **medicine**, and **industry**.



**ESSENTIAL
QUESTION**

How do Pascal's Principle and Archimedes' Principle improve our daily lives?



Easy Way to Remember

- **Archimedes = Afloat** → Think "**A**" for **Archimedes** and "**A**" for **Afloat** (floating and buoyancy).
- **Pascal = Pressure** → Think "**P**" for **Pascal** and "**P**" for **Pressure** (hydraulic systems).

Summary

- **Archimedes' Principle:** Deals with **buoyancy** and explains **why objects float or sink**.
- **Pascal's Law:** Deals with **pressure** in confined fluids and explains **how hydraulic machines work**.